Editorial

Risk-Adapted Screening for Methicillin-Resistant *Staphylococcus aureus*

Frieder Schaumburg

Editorial to accompany the article: "Screening for Methicillin-Resistant *Staphylococcus aureus* —an Analysis Based on Findings From the Hospital Infection Surveillance System (KISS), 2006–2021" by Miriam Wiese-Posselt et al. in this issue of *Deutsches Ärzteblatt International*

Methicillin-resistant *Staphylococcus aureus* (MRSA) has been part of everyday clinical practice for many years now. Besides livestock-associated (LA) MRSA and community-acquired (CA) MRSA, hospital-associated (HA) MRSA is of special importance in the care of hospitalized patients.

MRSA infections are dreaded because they are associated with increased morbidity and mortality (1).

Declining trend

Since 1 July 2009, the detection of MRSA in blood cultures or cerebrospinal fluid has been subject to mandatory reporting pursuant to section 7 of the German Protection against Infection Act (IfSG, Infektionsschutzgesetz). Over the past decade, there has been a steady decrease in the absolute number of reported invasive MRSA infections (2012: 4508 cases; 2022: 1040 cases) (2). A similar trend is also evident in the drug resistance findings of the Robert Koch-Institute's antibiotic resistance surveillance (3).

It is likely that the decrease in MRSA infections can be attributed to a number of factors. The fitness of currently circulating MRSA clones is declining (4) and many measures and bundles of measures have effectively curbed the spread of MRSA (for example, hygiene measures, antibiotic-stewardship programs, availability of antibiotics which are effective against MRSA). It seems that the disease has lost some of its dread.

Significance of screening frequency

In all areas of medicine, it is imperative to critically challenge established procedures on a regular basis. Often, large datasets are required to evaluate the evidence in support of specific measures. In Germany, the Hospital Infection Surveillance System (KISS, Krankenhaus-Infektions-Surveillance-System) provides a database in which (nosocomial) infections are systematically recorded by the participating hospitals (5).

In a retrospective study, Wiese-Posselt et al. analyzed KISS data (MRSA-KISS module) to quantify the incidence and prevalence of MRSA cases in the period 2006–2021 and to determine whether the MRSA screening frequency is associated with the incidence of nosocomial MRSA cases (6). Here, the underlying question is whether MRSA admission screening has in fact prevented MRSA cases.

The authors first confirm that there is a decrease in the prevalence of MRSA among patients. As expected, hospitals with a high screening frequency (nasal swabs for MRSA/100 patient admissions) also had a higher MRSA admission prevalence, as more cases were detected due to intensified screening. The screening frequency expresses the proportion of patients who are screened for MRSA colonization on admission. The screening frequency of hospitals that only screen patients with MRSA risk factors is typically about 40% (7). In other words, 40% of patients are tested for MRSA upon admission.

Nosocomial incidence density

Nosocomial incidence density (nosocomial MRSA cases/1000 patient days) also declined over the observation period. Nosocomial incidence density is commonly used as a measure to determine the risk of MRSA infection during a hospital stay. Here, MRSA cases are recorded over time, together representing all patients "at risk", i.e. staying in the hospital. Rather than considering individual hospital stays, the length of hospital stays of individual patients is added up.

An interesting point in the analysis by Wiese-Posselt et al. is that the nosocomial incidence density was comparable between hospitals, regardless of screening frequency. In other words, a high screening frequency was not associated with a low incidence density. The study did not address the question of whether the incidence density was also reduced by other, possibly compensatory measures (for example, improved hygiene measures).

Consequences of MRSA colonization

If MRSA is detected during admission screening, it usually has far-reaching consequences. These include, among others:

- Accommodation of the patient in a single room
- Personal protective equipment used by clinical staff when entering the patient room (wearing of gowns, gloves)

Institute of Medical Microbiology, University Hospital Münster (UKM), Münster, Germany: Prof. Dr. med. Frieder Schaumburg

- Adaption of perioperative antibiotic prophylaxis
- Considering MRSA in calculated antibiotic therapy
- Decolonization with mupirocin, and
- Antiseptic treatments.

Conclusion

The study by Wiese-Posselt et al. will add to the discussion on the effectiveness of MRSA screening, despite methodological limitations. With few exceptions, there have not yet been any prospective, (cluster-) randomized intervention studies to evaluate the effectiveness of hygiene measures. For this reason, and in line with the conclusion of Wiese-Posselt et al., it is clear that the current recommendation of the Commission for Hospital Hygiene and Infection Prevention (KRINKO) remains fully valid. That means: At the very least, patients at risk must be screened for MRSA colonization upon admission to the hospital (8).

Conflict of interest statement
The author declares that no conflict of interest exists.

Manuscript received on 17 May 2023, revised version accepted on 17 May 2023.

References

1. Bai AD, Lo CKL, Komorowski AS, et al.: Staphylococcus aureus bacteraemia mortality: a systematic review and meta-analysis. Clin Microbiol Infect 2022; 28: 1076–84.
2. Robert Koch-Institut: SurvStat@RKI 2.0. https://survstat.rki.de/ (last accessed on 10 May 2023).
3. Robert Koch-Institut: ARS – Antibiotika-Resistenz-Surveillance. https://ars.rki.de/ (last accessed on 15 May 2023).
4. Touati A, Bellil Z, Barache D, Mairi A: Fitness cost of antibiotic resistance in staphylococcus aureus: a systematic review. Microbial drug resistance (Larchmont, NY) 2021; 27: 1218–31.
5. Kerwat K, Geffers C, Gastmeier P, Wulf H: [The hospital infection surveillance system (KISS)]. Anasthesiol Intensivmed Notfallmed Schmerzther 2010; 45: 562–3.
6. Wiese-Posselt M, Saydan S, Schwab F, et al.: Screening for methicillin-resistant Staphylococcus aureus—an analysis based on findings from the Hospital Infection Surveillance System (KISS), 2006–2021. Dtsch Arztebl Int 2023; 120: 447–53.
7. Robert Koch-Institut: Zum Aufwand von MRSA-Screeninguntersuchungen in deutschen Krankenhäusern. Epidemiologisches Bulletin 2013; 5.
8. Ruscher C: Empfehlungen zur Prävention und Kontrolle von Methicillin-resistenten Staphylococcus aureus-Stämmen (MRSA) in medizinischen und pflegerischen Einrichtungen. Bundesgesundheitsblatt Gesundheitsforschung Gesundheitsschutz 2014; 57: 695–732.

Corresponding author
Prof. Dr. med. Frieder Schaumburg
Institut für Medizinische Mikrobiologie
Universitätsklinikum Münster (UKM)
Domagkstraße 10, 48149 Münster, Germany
frieder.schaumburg@ukmuenster.de

Cite this as:
Schaumburg F:
Risk-adapted screening for methicillin-resistant *Staphylococcus aureus.*
Dtsch Arztebl Int 2023; 120: 445–6. DOI: 10.3238/arztebl.m2023.0131

CLINICAL SNAPSHOT

Genital Herpes Zoster in Childhood

An 11-year-old girl presented to the emergency department reporting a 3-day history of mildly painful blisters extending from the region of the left labia to the left perineum as well as dysuria since the previous day. Physical examination revealed partially crusted, vesicular eruptions on an erythematous base *(Figure)* in addition to swollen lymph nodes in the groin. Upon subsequent gynecological examination, a diagnosis of genital herpes and, thus, sexual contact or abuse was suspected. The patient repeatedly rejected this suggestion and was presented to our pediatric department. Due to the unilaterality of lesions, pain, and varicella infection 4 years previously, polymerase chain reaction (PCR) testing for varicella was performed on secretion from the blisters. This was positive. Detection of herpes simplex 1 or 2 was unsuccessful.
Genital herpes zoster is a very rare differential diagnosis in childhood and is detected by means of PCR from vesicular secretions. It is not possible to differentiate between herpes simplex and herpes zoster on the basis of clinical presentation. In the case of unilateral eruptions in the genital region, herpes zoster should be considered in the differential diagnosis of genital herpes irrespective of age.

Dr. med. Sophie Jocham, Dr. med. Peter Küster, Ludgerusklinken Münster, Clemenshospital, Kinderheilkunde, sophie.jocham@googlemail.com

Conflict of interest statement: The authors declare that no conflict of interest exists.

Translated from the original German by Christine Rye.

Cite this as: Jocham S, Küster P: Genital herpes zoster in childhood. Dtsch Arztebl Int 2023; 120: 446. DOI: 10.3238/arztebl.m2023.0042